



ADITYA

COMPUTER ACADEMY

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FUNDAMENTAL NOTES

NAME :-

ADDRESS :-

MOB. NO:-

BATCH TIME :-

Computer Fundamental Notes (English)

What is a Computer?

A computer is an electronic machine that accepts data as input, processes it, and provides useful information as output. It can perform calculations, store data, and manage different tasks quickly and accurately.

Features of Computer

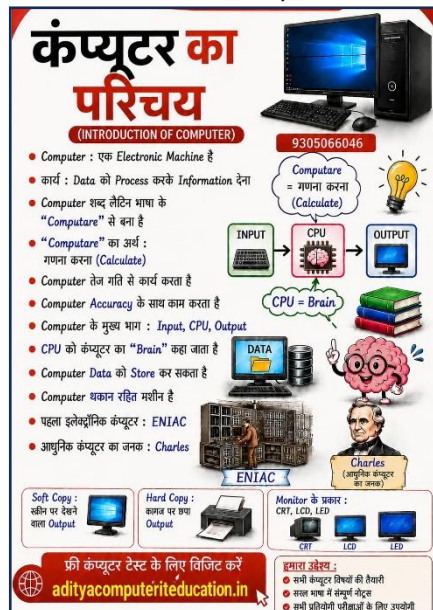
- High Speed
- Accuracy
- Large Storage Capacity
- Works Without Tiredness
- Multitasking

Father of Computer

Charles Babbage is known as the “Father of Computer.” He designed the Analytical Engine in 1837.

Full Form of COMPUTER

- C – Commonly
- O – Operating
- M – Machine
- P – Particularly
- U – Used For
- T – Technology
- E – Educational And
- R – Research



Types of Computer

1. Analog Computer

An Analog Computer works on continuously changing physical quantities like temperature, pressure, speed, and electric current. It uses signals instead of numbers.

Features

- Works on Continuous Data
- Gives Fast Results
- Used in Scientific and Medical Fields
- Less Accurate than Digital Computer

Examples

- Speedometer
- Thermometer
- ECG Machine
- Petrol Pump Meter

2. Digital Computer

A Digital Computer processes data in the form of digits (0 and 1). It works on the Binary Number System. These are the most commonly used computers today.

Features

- Very Fast
- High Accuracy
- Can Store Data
- Performs Logical and Arithmetic Operations
- Easy to Use



Examples

- Desktop Computer
- Laptop
- Calculator
- Smartphone
- ATM Machine

3. Hybrid Computer

A Hybrid Computer combines the features of Analog and Digital Computers. It can process both continuous and digital data.

Features

- Very Fast
- Processes Both Types of Data
- Gives Accurate Results
- Used in Scientific and Medical Fields

Examples

- ICU Monitoring Machine
- ECG Machine
- Petrol Pump Machine
- Weather Forecasting System

TYPES OF COMPUTER (BASED ON SIZE)

1. Micro Computer

Small-sized computers.

Examples: Desktop, Laptop

2. Mini Computer

Medium-sized computers used in small organizations.

3. Mainframe Computer

Large computers where thousands of users can work together.

Example: Banking Systems

4. Super Computer

The fastest and most powerful computer.

Used for weather forecasting and scientific research.

GENERATIONS OF COMPUTER

Generation	Time Period	Technology Used	Main Feature
First Generation	1940–1956	Vacuum Tube	Large Size
Second Generation	1956–1963	Transistor	Faster Speed
Third Generation	1964–1971	IC	Less Electricity Consumption
Fourth Generation	1971–Present	Microprocessor	Personal Computer
Fifth Generation	Present & Future	Artificial Intelligence	Smart Technology

First Generation Computer

Time Period

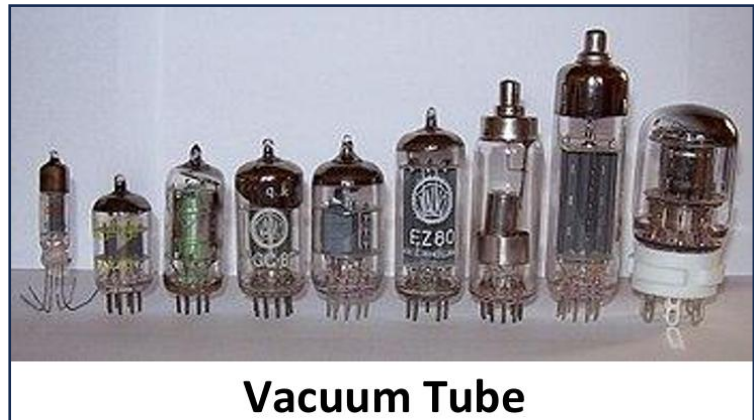
1940–1956

Technology Used

Vacuum Tube

Features

- Very Large Size
- High Electricity Consumption
- Produced More Heat
- Slow Speed
- Used Machine Language
- Difficult Maintenance



Examples

- ENIAC
- EDVAC
- UNIVAC I



Second Generation Computer

Time Period

1956–1963

Technology Used

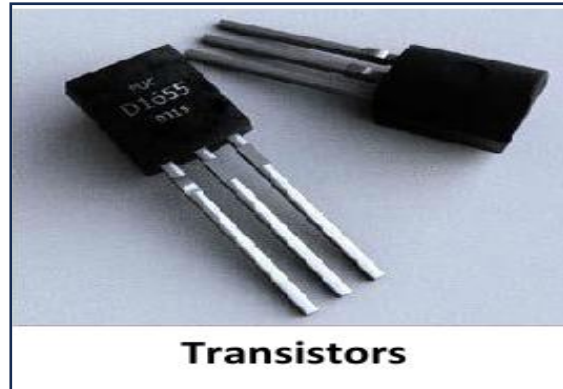
Transistor

Features

- Smaller than First Generation
- Less Electricity Consumption
- Faster Speed
- Less Heat
- Used Assembly Language
- Easier Maintenance

Examples

- IBM 1401
 - IBM 7094
 - CDC 1604
-



Transistors

Third Generation Computer

Time Period

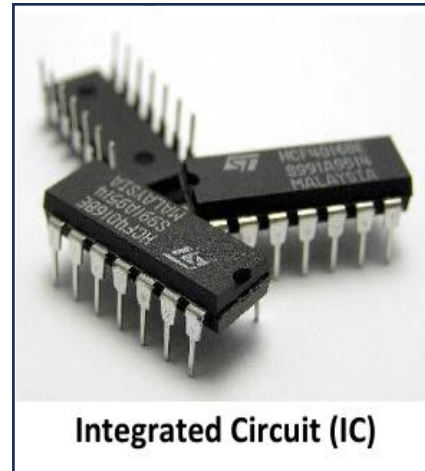
1964–1971

Technology Used

Integrated Circuit (IC)

Features

- Smaller Size
 - Faster Processing
 - Better Reliability
 - Lower Power Consumption
-



Fourth Generation Computer

Time Period

1971–Present

Technology Used

Microprocessor

Features

- Very Small Size
- Very Fast Speed
- Low Power Consumption
- Lower Cost
- Development of Personal Computers
- Use of GUI (Graphical User Interface)

Examples

- IBM PC
- Apple Macintosh



Fifth Generation Computer

Technology Used

Artificial Intelligence (AI)

Features

- Artificial Intelligence
- Very Fast Processing
- Voice Recognition
- Automatic Decision Making
- Natural Language Processing
- Used in Internet and Robotics

Examples

- ChatGPT
- Google Assistant
- AI Robots
- Super Computers

Hardware

Hardware refers to the physical parts of a computer that we can touch.

Examples

- Keyboard
- Mouse
- Monitor
- Printer
- CPU



Software

Software is a program that gives instructions to the computer to perform tasks. We cannot touch software; we can only use it.

Examples

- Microsoft Windows
 - Microsoft Word
-

Types of Software

1. System Software

System Software controls the computer hardware and helps the computer work properly. It acts as an interface between the user and hardware.

Operating System

An Operating System is the main software of a computer.

Functions

- Memory Management
- File Management
- Device Control

Examples

- Microsoft Windows
 - Linux
 - Android
-

2. Application Software

Application Software is used by users according to their needs.

Examples

- Microsoft Word
 - Microsoft Excel
 - WhatsApp
 - Facebook
-

3. Utility Software

Utility Software is used for maintenance, security, and improving computer performance. It helps remove viruses, manage files, and increase system speed.

Examples

- Antivirus
- Disk Cleanup
- Backup Software
- File Compression Software

Simple Definition

“Software that keeps the computer safe and fast is called Utility Software.”

Input Device

Devices used to enter data into the computer are called Input Devices.

Examples

- Keyboard
 - Mouse
 - Scanner
 - Microphone
-

Output Device

Devices that show the result from the computer are called Output Devices.

Examples

- Monitor
 - Printer
 - Speaker
 - Projector
-

Computer Memory

Memory is the place where data and instructions are stored.

Types of Memory

A. Primary Memory

This is the main memory of the computer.

1. RAM (Random Access Memory)

- Temporary Memory
- Data is lost when power is off
- Fast Memory

2. ROM (Read Only Memory)

- Permanent Memory
 - Data remains stored permanently
 - Data can only be read
-

B. Secondary Memory

Used for permanent storage.

Examples

- Hard Disk
 - Pen Drive
 - CD/DVD
 - Memory Card
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First Use of Computer in India

The first computer in India was used at the Indian Statistical Institute in Kolkata.

Main Parts of Computer

The computer is made up of several important parts.

Main Parts

1. Keyboard
 2. Mouse
 3. CPU
 4. Monitor
 5. Printer
-

CPU (Central Processing Unit)

CPU is the most important part of the computer. It is called the “Brain of the Computer.”

Functions

- Processes Data
- Performs Calculations
- Controls All Parts of the Computer

Main Parts of CPU

1. ALU
 2. CU
 3. Memory Unit
-



Monitor

Monitor is an Output Device that displays processed information on the screen.

Uses

- Watching Videos
- Doing Work
- Viewing Results

Types of Monitor

1. CRT Monitor
2. LCD Monitor
3. LED Monitor
4. Touch Screen Monitor



Keyboard

Keyboard is an Input Device used to type letters, numbers, and commands into the computer.

Uses

- Typing
- Giving Commands
- Entering Data

Types of Keyboard

1. Wired Keyboard
 2. Wireless Keyboard
 3. Mechanical Keyboard
 4. Multimedia Keyboard
-

Mouse

Mouse is an Input Device used to control the pointer on the computer screen.

Uses

- Clicking
- Opening Files
- Drag and Drop

Types of Mouse

1. Wired Mouse
2. Wireless Mouse
3. Optical Mouse
4. Laser Mouse